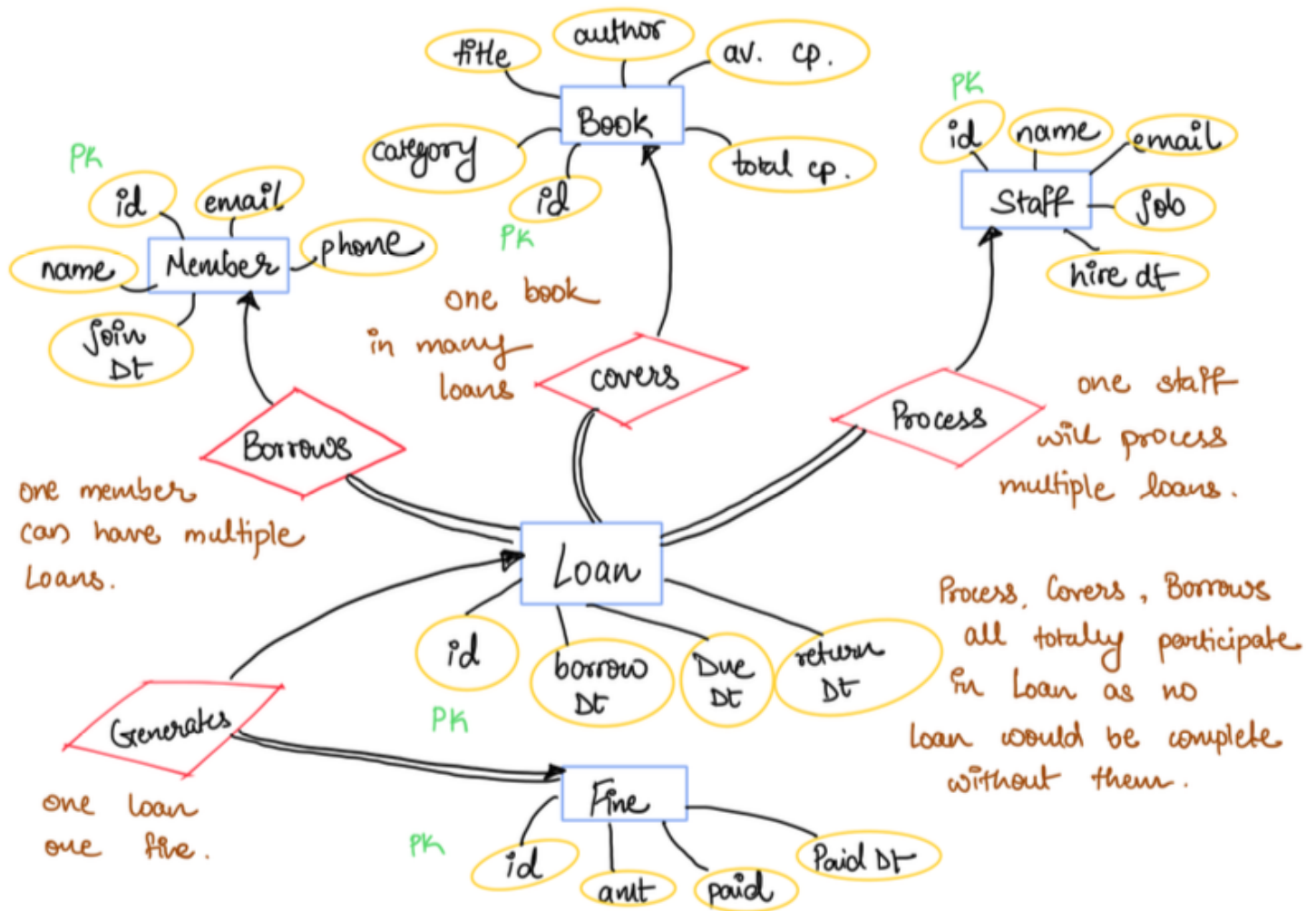


Phase 2: ER Diagram



Reasoning

I designed this ER diagram using five entity sets in this diagram. Member, Book, Staff, and Fine each store attributes that describe a real-world object on their own. Loan is also modeled as its own entity rather than a simple relationship diamond, since it needs to participate in a relationship with Fine in addition to Member, Book, and Staff. This would not be possible if Loan were only a diamond, since diamonds in this notation connect entities to each other but do not themselves connect to other relationships.

There are four relationship sets, all binary. Borrows connects Member and Loan, with the arrow pointing toward Member since one member can have many loans, and a double line from Loan showing every loan must belong to a member. Covers and Process follow the same pattern for Book and Staff. Generates connects Loan and Fine as a one-to-one relationship, with a double line from Fine, since a fine cannot exist without a loan, and a single line from Loan, since most loans never generate a fine.

I intentionally left Staff and Book unconnected, since they never interact directly in this system. Their only relationship is indirect, through the loans that a staff member processes and the books those loans cover.

This diagram sets up the schema design phase directly. Every primary key shown here becomes a table's primary key, and every relationship with an arrow pointing toward the "one" side tells me exactly where a foreign key needs to be placed once I convert this into relational tables.